

APM1110 - FA1

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GitHub Link

1. Write the skewness program, and use it to calculate the skewness coefficient of the four examination subjects in results.txt (results.csv). What can you say about these data?

```
results <- read.csv("results.csv")

skewness <- function(x) {
  xbar <- mean(is.na(x))
  sum2 <- sum((x - xbar) ^ 2, na.rm = T)
  sum3 <- sum((x - xbar) ^ 3, na.rm = T)
  skew <- (sqrt(length(x)) * sum3 / (sum2 ^ (1.5)))
}

skew1 <- c(
  arch1 = skewness(results$arch1),
  prog1 = skewness(results$prog1),
  arch2 = skewness(results$arch2),
  prog2 = skewness(results$prog2)
)

skew1
```

```
##      arch1      prog1      arch2      prog2
## 1.167807 1.171101 1.248553 1.272761
```

The four results show positive skewness which indicates that the distributions are right-skewed, stating that they have a longer tail toward higher scores. The arch2 and prog2 are greater than arch1 and prog1. This data generally presents that the results are not exactly symmetric.

Pearson has given an approximate formula for the skewness that is easier to calculate than the exact formula given in Equation 2.1: $\text{skew} = (3 (\text{mean} - \text{median})) / \text{standard deviation}$. Write a program to calculate this and apply it to the data in results.txt (results.csv). Is it a reasonable approximation?

```
results <- read.csv("results.csv")

skew <- function(x) {
  3 * (mean(x, na.rm = TRUE) - median(x, na.rm = TRUE)) /
  sd(x, na.rm = TRUE)
}
```

```
skew2 <- c(
  arch1 = skew(results$arch1),
  prog1 = skew(results$prog1),
  arch2 = skew(results$arch2),
  prog2 = skew(results$prog2)
)
```

```
skew2
```

```
##      arch1      prog1      arch2      prog2
## -0.5896811 -0.6345275  0.5190494 -0.3854051
```

Pearson's approximate formula does not appear to be a reasonable approximation for these data. The skewness values vary significantly between the two formulas. One of them is more than 1 while the other sets are all negative. Therefore, Pearson's are limited to data with strong median, which is known as the Second Coefficient.

2. For the class of 50 students of computing detailed in Exercise 1.1, use R to

(a) form the stem-and-leaf display for each gender, and discuss the advantages of this representation compared to the traditional histogram;

```
f <- c(57, 59, 78, 79, 60, 65, 68, 71, 75, 48, 51, 55, 56, 41, 43,
      44, 75, 78, 80, 81, 83, 83, 85)
```

```
m <- c(48, 49, 49, 30, 30, 31, 32, 35, 37, 41, 86, 42, 51, 53, 56,
      42, 44, 50, 51, 65, 67, 51, 56, 58, 64, 64, 75)
```

```
stem(f)
```

```
##
## The decimal point is 1 digit(s) to the right of the |
##
##  4 | 1348
##  5 | 15679
##  6 | 058
##  7 | 155889
##  8 | 01335
```

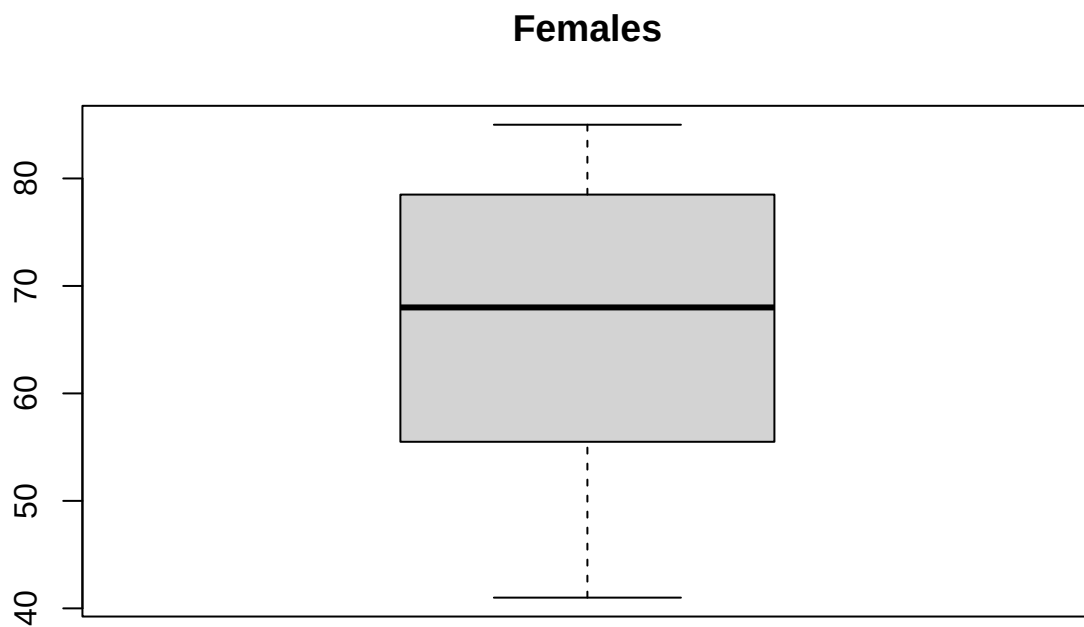
```
stem(m)
```

```
##
## The decimal point is 1 digit(s) to the right of the |
##
##  3 | 001257
##  4 | 1224899
##  5 | 01113668
##  6 | 4457
##  7 | 5
##  8 | 6
```

Stem-and-leaf displays are used to organize numerical data rapidly. It represents the whole shape and the spread of distribution, useful for smaller datasets. It provides an immediate arranged view in a single glance compared to a histogram's unreadable and cluttered data.

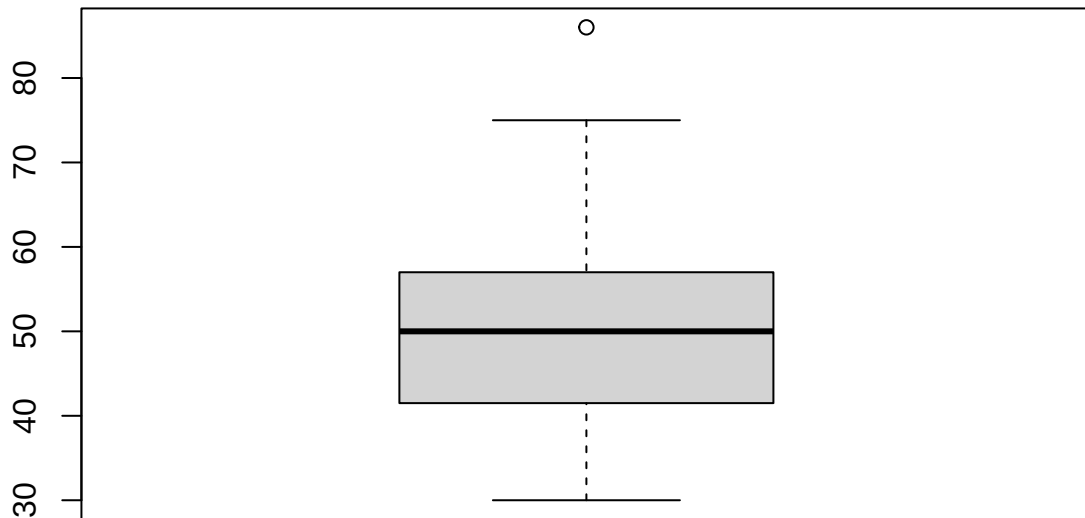
(b) construct a box-plot for each gender and discuss the findings.

```
f <- c(57, 59, 78, 79, 60, 65, 68, 71, 75, 48, 51, 55, 56, 41, 43,  
      44, 75, 78, 80, 81, 83, 83, 85)  
  
m <- c(48, 49, 49, 30, 30, 31, 32, 35, 37, 41, 86, 42, 51, 53, 56,  
      42, 44, 50, 51, 65, 67, 51, 56, 58, 64, 64, 75)  
  
boxplot(f,  
        main = "Females")
```



```
boxplot(m,  
        main = "Males")
```

Males



The thickest line in the middle of the box is the median, which is 68 for females and 50 for males. The grey bar below the median line is the 1st quartile, which starts from 55.5 in females' and 41.4 in males'. The upper portion from the median is the 3rd quartile, ending in 78.5 for females and 57 for males. The whole grey box represents the middle 50% of the datapoints, and underneath it, the lower whisker, a thinner line, is the lowest point from the data. Opposite to that side is the highest score attained. The females' lowest score is 41 and the highest is 85, while the males' lowest is 30 and the highest is 75. But there is a single dot exceeding the upper whisker, a value of 86 (which is the highest attained mark of a male student), meaning it lies too far from the Inter Quartile Range ($1.5 * IQR$).